

R&D data source unification

A candidate-review assistant combining field observations, operations, laboratory results and genomics. Users inspect assessments, open evidence and record a reasoned ADVANCE, HOLD or DISCARD decision.

PROBLEM & SOLUTION Senior breeders need evidence spread across files and systems. One browser workflow brings the candidate assessment, supporting source rows and decision record together.	HUMAN DECISION The demo uses SME-supplied synthetic data for 150 candidates. Assessments are provisional; the breeder can challenge them and makes the final choice. This is decision support, not automated breeding.
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65 RED Fails a rule threshold	53 AMBER Needs further review	32 GREEN Meets the rule criteria
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Baseline matches all 150 supplied synthetic recommendations. This establishes output consistency, not biological validation.

01 / TECHNICAL APPROACH

Calculation separate from explanation. Deterministic Python reconstructs evidence and applies a versioned, provisional policy. Assessments include criteria, warnings and reasons.

Traceable history. Each saved decision preserves the recommendation and evidence available at that time. Reviewed corrections create new revisions without replacing earlier records.

Grounded AI, explicit actions. Read-only MCP tools supply evidence for cited LLM answers. Natural-language filters return an editable preview requiring Apply. Neither AI workflow records decisions.

Recorded verification. 473 offline tests, 43 candidate browser tests and extracted-package startup/persistence checks. Setup, API and process guides support handover. [Verification](#) · [Handover](#)

02 / IMPLEMENTATION STACK

LAYER	TECHNOLOGY	USE / RATIONALE
Data / rules	Python 3.12+ · pandas · NumPy	Load CSV/ZIP tables, validate joins and calculate repeatable assessments.
Backend	FastAPI · Pydantic · Uvicorn	Serve HTTP APIs and the dashboard; validate inputs.
Interface	HTML · CSS · JavaScript	Browser-based review without a frontend framework or build step.
Database	SQLite · SHA-256 snapshots	Transactional history and saved source versions; no database server.
AI / tools	MCP · Portkey · OpenAI SDK	Read-only evidence retrieval and cited answers through a configurable model route.
Testing	uv · pytest · Playwright	Locked dependencies, logic/API tests and browser checks.

Supporting tools: AnyIO + httpx2 (async calls); localStorage (preferences); dotenv + TOML (configuration); Matplotlib (analysis); Git + ZIP launchers (delivery). Legacy logs use JSONL.

03 / PLANNED LIVE DEMO

AMBER CANDIDATE Select a candidate, inspect its criteria and source rows, ask the live LLM to explain the assessment, then open a citation. Live AI requires configured credentials and an available model route.	REVIEW AND RECORD HOLD Manually review the evidence, confirm HOLD with a reason, then reopen the saved decision and its original evidence to demonstrate traceability.
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04 / IMPACT & SCALABILITY

Intended benefit: less cross-file lookup and a clearer decision record. Browser review requires no coding; installation and AI setup remain prerequisites. General time savings and productivity gains have not been measured.

Syngenta Pull: a breeder-reviewed pilot with authorized crop data and SME-approved scoring. **HatchWorks Leverage:** reuse the separation of deterministic assessments, grounded explanations and human decisions.

AI supported planning, data analysis, implementation, testing, documentation and rehearsal. Humans selected scope and approved changes. The [prompt log](#) retains failures, corrections and deferred work.

Lesson: arrange approved tools, access instructions and repository setup before the event. Tool-access difficulties are team observations, not verified organization-wide findings.

05 / TEAM & GENDD

06 / OPERATING LIMITS

Local prototype, not a proven hosted multi-user service. No live research connections; incomplete pedigree and self-declared identities. Scaling needs validated policies, production identity/permissions and reliable model access. Manual evidence review and decisions work without AI credentials.

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